

Computer Networks — Placement Notes

High-yield revision guide for software engineering placements, technical interviews, MCQs, and viva

1. Network Models: OSI and TCP/IP

OSI has 7 layers: Physical, Data Link, Network, Transport, Session, Presentation, Application. TCP/IP commonly uses Link, Internet, Transport, Application.

Layer	Main responsibility	Examples / key terms
Application	Services used by applications	HTTP, HTTPS, FTP, SMTP, DNS, DHCP, SNMP
Presentation	Syntax, format, encryption/compression	Encoding, serialization, encryption
Session	Establish/manage sessions	Dialog control, token management, synchronization
Transport	Process-to-process logical communication	TCP, UDP, ports, reliability, flow/congestion control
Network	Host-to-host delivery and routing	IP, ICMP, routing, routers
Data Link	Node-to-node frames on a link	Frames, MAC, switches, error detection
Physical	Raw bit transmission	Signals, media, topology

Interview: Which devices operate where? Typical interview answer: hub/repeater → Physical; switch/bridge → Data Link; router → Network.

2. TCP vs UDP — Must Know

Feature	TCP	UDP
Connection	Connection-oriented; handshake	Connectionless; no handshake
Reliability	Reliable, ordered, no duplicates	Best effort; no delivery/order guarantee
Data model	Byte stream	Datagrams/messages
Control	Flow + congestion control	No built-in flow/congestion control
Header	Larger (typically 20+ bytes)	8 bytes
Use cases	HTTP/HTTPS, FTP, SMTP, SSH	DNS commonly, streaming/real-time, DHCP, NTP

TCP socket identification: 4-tuple — source IP, source port, destination IP, destination port. **UDP socket demultiplexing:** commonly identified by destination IP and destination port.

Multiplexing: sender gathers data from multiple sockets, adds transport headers, sends segments downward.

Demultiplexing: receiver delivers incoming segment data to the correct socket.

Interview: Why is UDP faster/lighter? No connection setup, smaller header, and no built-in retransmission, ordering, flow control, or congestion control.

3. Ports and Common Protocols

A port is a logical communication endpoint represented by a 16-bit number (0–65535). Ports 0–1023 are well-known/reserved ports.

Protocol	Default port	Transport	Purpose
HTTP	80	TCP	Web
HTTPS	443	TCP	Secure web (HTTP + TLS)
FTP control	21	TCP	FTP commands/control
SMTP	25	TCP	Mail transfer between servers

Protocol	Default port	Transport	Purpose
POP3	110	TCP	Mail retrieval
DNS	53	UDP commonly; TCP also	Name resolution
DHCP	67/68	UDP	Dynamic network configuration
SNMP	161/162	UDP	Network monitoring/management
RIP	520	UDP	Distance-vector routing

4. HTTP / HTTPS

HTTP is an application-layer, client-server protocol. It is **stateless**: the server does not inherently remember previous client requests. HTTP traditionally runs over TCP. HTTPS is HTTP protected using TLS, commonly on port 443.

HTTP request essentials

- **Method** such as GET/POST; request target/path; HTTP version; headers; optional body.
- **Common status codes**: 200 OK, 301 Moved Permanently, 400 Bad Request, 404 Not Found, 505 HTTP Version Not Supported.

Interview: Why is HTTP called stateless? Each request can be handled independently; protocol-level server behavior does not require remembering earlier requests. State can still be implemented using cookies, sessions, tokens, etc.

5. FTP

FTP is an application-layer file-transfer protocol over TCP. It uses separate connections: a **control connection** and a **data connection**. The control connection commonly uses server port 21 and remains for the session; data connections carry file/listing data.

- **Out-of-band control**: FTP separates control information from transferred data.
- Commands: USER, PASS, LIST, RETR filename (download), STOR filename (upload).
- FTP maintains session state such as authenticated user and current remote directory.

Interview: FTP vs HTTP control? FTP uses separate control/data connections (out-of-band); HTTP carries headers/control and content through the HTTP connection (in-band).

6. E-mail: SMTP, POP3, IMAP

SMTP transfers outgoing mail and mail between mail servers using TCP; it is primarily a **push** protocol and uses persistent connections. Traditional server port: 25.

POP3 is a simple retrieval protocol (port 110). Typical phases: authorization, transaction, update. **IMAP** keeps mail organized on the server, supports folders/search, and maintains user state across sessions.

HTTP vs SMTP: HTTP is mainly pull; SMTP is push. SMTP historically required 7-bit ASCII transfer representation; MIME enables richer content. HTTP typically returns objects as separate responses, while an e-mail message can contain multiple MIME parts.

Interview: Webmail uses what? Browser ↔ mail service commonly uses HTTP/HTTPS; mail servers still use SMTP for server-to-server transfer.

7. DNS — Very Important

DNS (Domain Name System) maps hostnames to IP addresses using a distributed hierarchical naming system.

- Services: hostname → IP translation, host aliasing, mail-server aliasing (MX), and load distribution.
- DNS queries commonly use UDP port 53; TCP is also used when needed, including some large responses and zone transfers.

- DNS can rotate/order multiple IP addresses associated with a hostname to distribute requests.

Interview: What happens before opening a website if IP is unknown? The client/resolver performs DNS resolution to obtain an IP, then networking/transport connection setup proceeds.

8. DHCP, SNMP, NTP

- **DHCP:** automatically supplies network configuration such as IP address, subnet mask, default gateway/router, and DNS server. Uses UDP.
- **SNMP:** monitors/manages network devices; UDP ports 161/162 are commonly used.
- **NTP:** synchronizes clocks over a network; commonly uses UDP.

Interview: Why does DHCP use broadcast initially? A new client may not yet know its own IP or the DHCP server's IP, so discovery can begin using broadcast communication.

9. Transport Layer Fundamentals

The transport layer provides **logical process-to-process communication** between applications on end systems. Transport protocols are implemented primarily at end hosts, not intermediate routers.

- TCP: reliable, connection-oriented.
- UDP: unreliable/best-effort, connectionless.
- Transport packet terminology: TCP segment; UDP datagram is common terminology, though 'segment' may be used generically in teaching.

Interview: Network vs Transport layer? Network layer provides host-to-host delivery/routing; transport layer provides process-to-process communication.

10. UDP Header and Checksum

UDP header has four 16-bit fields: **source port, destination port, length, checksum**. Length includes header + data.

Checksum uses one's-complement arithmetic over 16-bit words (including a pseudo-header in IP usage). At the receiver, checksum verification helps detect corruption. UDP itself does not provide retransmission, ordering, or flow control.

Interview: When choose UDP? Low overhead, latency-sensitive real-time traffic, simple request/response such as many DNS queries, or multicast/broadcast scenarios.

11. TCP Connection and Buffers

TCP provides a **full-duplex, point-to-point** connection. Each endpoint maintains connection state including send/receive buffers and variables. Routers in between do not maintain TCP connection buffers.

MSS (Maximum Segment Size) is the maximum TCP application-data payload in a segment, not the whole segment. It is selected with MTU/header overhead in mind to avoid fragmentation.

Interview: MTU vs MSS? MTU limits the size of a link-layer payload/IP packet carried on a link; MSS limits TCP payload data per segment.

12. TCP Header — Interview Favorite

Field	Purpose
Source/Destination ports	Process/socket addressing
Sequence number	Byte number of first data byte in this segment
Acknowledgment number	Next byte expected from peer
Header length	TCP header size
Receive window	Flow control

Field	Purpose
Checksum	Error detection
Flags	SYN, ACK, FIN, RST, PSH, URG (modern headers also define others)
Options	MSS, window scaling, timestamps, etc.

13. Sequence Numbers and ACKs

TCP treats application data as an ordered **byte stream**. A segment's sequence number is the number of its first data byte. If MSS=1000 and first byte number=0, segment sequence numbers may be 0, 1000, 2000, ...

Acknowledgment number = next byte expected. If bytes 0–535 arrived correctly, ACK=536. TCP uses **cumulative acknowledgments**: if bytes 536–899 are missing but later bytes arrived, the ACK still indicates 536.

Each side chooses an initial sequence number (ISN), rather than always starting at zero.

Interview: If receiver sends ACK 5000, what does it mean? It has cumulatively received bytes through 4999 in order and expects byte 5000 next.

14. TCP Three-Way Handshake

Step	Client/Server action	Key values
1. SYN	Client requests connection	SYN=1, Seq=x
2. SYN-ACK	Server accepts and synchronizes	SYN=1, ACK=1, Seq=y, Ack=x+1
3. ACK	Client acknowledges server	ACK=1, Ack=y+1

After the handshake, both sides have synchronized initial sequence numbers and established connection state. SYN flooding exploits resource allocation associated with many incomplete connection attempts.

Interview: Why three-way, not two-way? Both sides must confirm reachability and synchronize/acknowledge each other's initial sequence numbers, including confirmation that the server's response reached the client.

15. RTT and TCP Timeout

SampleRTT is measured from segment transmission to acknowledgment receipt (normally avoiding retransmitted samples due to ambiguity). TCP smooths measurements using an exponentially weighted moving average.

EstimatedRTT = $0.875 \times \text{EstimatedRTT} + 0.125 \times \text{SampleRTT}$

DevRTT = $0.75 \times \text{DevRTT} + 0.25 \times |\text{SampleRTT} - \text{EstimatedRTT}|$

TimeoutInterval = $\text{EstimatedRTT} + 4 \times \text{DevRTT}$

On timeout, retransmission timeout is typically backed off (for example, doubled) to reduce repeated premature retransmissions.

Interview: Why not use only latest SampleRTT? Network delay fluctuates; smoothing provides a more stable timeout estimate.

16. TCP Flow Control

Flow control prevents a fast sender from overwhelming a receiver. The receiver advertises available buffer space as the **receive window (rwnd)**. The sender limits outstanding data accordingly.

Outstanding data $\leq \min(\text{cwnd}, \text{rwnd})$, where cwnd is congestion window and rwnd is receive window.

If rwnd becomes zero, the sender must avoid normal bulk transmission; TCP uses mechanisms such as window probes/persist behavior so it can learn when the receiver's window opens again.

Interview: Flow control vs congestion control? Flow control protects the receiver; congestion control protects the network.

17. TCP Congestion Control

Congestion occurs when offered network load exceeds available resources, increasing delay, loss, and retransmissions.

- **cwnd:** congestion window limits how much unacknowledged data the sender injects based on network conditions.
- **Slow Start:** cwnd grows rapidly (approximately exponentially per RTT) until a threshold/loss signal.
- **Congestion Avoidance:** growth becomes more conservative (approximately linear/additive increase).
- **Congestion Detection:** loss/ECN signals cause reduction of sending rate; exact behavior depends on TCP congestion-control algorithm.

Leaky bucket: shapes output toward a fixed rate. **Token bucket:** tokens accumulate at rate R up to capacity B, allowing controlled bursts when tokens are available.

Interview: cwnd vs rwnd? cwnd is sender-side network congestion limit; rwnd is receiver-advertised buffer capacity.

18. Network Layer: Forwarding vs Routing

Forwarding moves a packet from a router's incoming interface to the correct outgoing interface. **Routing** determines end-to-end paths using routing algorithms/protocols.

Interview: Simple distinction: forwarding is a local per-router data-plane action; routing computes/selects paths in the control plane.

19. IP and ICMP

IP provides best-effort packet delivery. **ICMP** carries network control/error information associated with IP, including destination unreachable, time exceeded, and parameter-related errors.

Interview: What makes traceroute possible? It relies on TTL/hop-limit behavior and ICMP Time Exceeded responses (implementation details vary).

20. OSPF vs RIP

Feature	OSPF	RIP
Type	Link-state	Distance-vector
Metric	Cost (commonly bandwidth-derived/configured)	Hop count
Algorithm concept	Shortest Path First (Dijkstra)	Distance-vector/Bellman-Ford family
Max hop limit	Not based on RIP's 15-hop limit	15 usable; 16 = unreachable
Other	DR/BDR election on eligible multi-access networks	Uses UDP port 520 (RIPv1/v2)

OSPF DR election: highest OSPF interface priority wins; ties are broken by highest Router ID. Router ID selection/configuration rules matter more than simply saying 'highest IP'.

Interview: Why does RIP not scale well? Hop-count metric is simplistic, convergence can be slower, and maximum usable path length is only 15 hops.

21. Data Link and Physical Layer

Data Link: organizes raw transmission into frames and handles link-level delivery/error-detection responsibilities. Ethernet switching and MAC addressing are core placement topics even though the source notes only briefly cover this layer.

Physical: transmits raw bits over media. Common topologies: mesh, bus, ring, star. Transmission modes: simplex, half-duplex, full-duplex.

Mode	Meaning	Example idea
Simplex	One direction only	One-way transmission
Half-duplex	Both directions, not simultaneously	Walkie-talkie
Full-duplex	Both directions simultaneously	Telephone/TCP logical communication

22. Peer-to-Peer and BitTorrent

P2P systems distribute work/data among peers. BitTorrent divides files into chunks/pieces and exchanges them among peers.

- **Rarest first:** request chunks with the fewest available copies among neighbors to improve availability.
- **Choking/unchoking:** prioritize peers providing good upload rates; periodically optimistically unchoke another peer to discover better partners and help newcomers.
- **Tit-for-tat incentive:** peers tend to reciprocate useful uploads.

DHT (Distributed Hash Table) distributes key-value lookup responsibility across participating nodes, avoiding reliance on one central lookup server.

23. What Happens When You Type google.com?

A strong interview answer should connect the layers in order:

- Browser parses URL and checks relevant caches/policies.
- DNS resolution obtains an IP address if not already cached.
- Host determines how to reach the destination; local-link delivery may require resolving the next-hop link-layer address.
- For HTTPS, a transport connection is established (classically TCP; modern HTTP/3 uses QUIC over UDP).
- For TCP, the three-way handshake establishes the connection.
- TLS handshake authenticates the server and establishes encryption keys for HTTPS.
- Browser sends an HTTP request; server/CDN/load balancer/application processes it.
- HTTP response returns status, headers, and content.
- Browser parses HTML, fetches dependent resources, executes scripts, performs layout/paint, and renders the page.

Interview: This question tests DNS, ARP/local-link delivery, routing, TCP/QUIC, TLS, HTTP, caching/CDN, and browser rendering. Explain in sequence, not as isolated definitions.

24. Rapid-Fire Placement Questions

Interview: TCP reliable how? — Sequence numbers, ACKs, retransmission, checksum/error detection, timers, ordered delivery, flow control; congestion control regulates network load.

Interview: TCP full-duplex? — Both endpoints can send and receive simultaneously over the same connection.

Interview: Why cumulative ACK? — ACK indicates the next in-order byte expected, implicitly acknowledging all earlier contiguous bytes.

Interview: HTTP stateless vs TCP stateful? — HTTP semantics do not inherently retain request history; TCP maintains connection state, sequence numbers, buffers, windows, etc.

Interview: DNS uses TCP or UDP? — Both. UDP is common for ordinary queries; TCP is used in cases such as zone transfers and when required by response/transport conditions.

Interview: What is a socket? — An OS abstraction/interface endpoint used by a process for network communication.

Interview: What is congestion? — Too much traffic for available network resources, causing queueing delay, drops, and reduced performance.

Interview: Flow vs error vs congestion control? — Flow protects receiver capacity; error/reliability mechanisms recover/detect corruption/loss; congestion control regulates load on network.

Interview: Router vs switch? — Router forwards between IP networks at Layer 3; a typical Ethernet switch forwards frames within LANs using Layer-2 information.

Interview: HTTPS? — HTTP over a secure transport using TLS, providing encryption, integrity, and authentication (primarily server authentication; client auth optional).

25. Last-Minute Memory Sheet

Topic	Remember
OSI	7 layers: Physical, Data Link, Network, Transport, Session, Presentation, Application
TCP/IP	Link, Internet, Transport, Application
TCP	Reliable, connection-oriented, ordered byte stream, full-duplex
UDP	Connectionless, low overhead, best effort
TCP handshake	SYN → SYN-ACK → ACK
TCP ACK	Next byte expected; cumulative
Flow control	rwnd protects receiver
Congestion control	cwnd protects network
DNS	Name resolution; port 53; UDP commonly + TCP when needed
HTTP / HTTPS	80 / 443
FTP	Control 21; separate control and data connections
SMTP / POP3	25 / 110
OSPF	Link-state; SPF/Dijkstra concept
RIP	Distance-vector; hop count; max 15 hops
Forwarding / Routing	Local packet movement / path determination

26. Corrections / Interview-Safe Clarifications

These notes preserve the important content from the supplied material while making a few statements safer for modern interviews:

- DNS is **not UDP-only**; it uses both UDP and TCP.
- HTTPS is best described as **HTTP secured with TLS**; modern protocol stacks may use HTTP/3 over QUIC rather than TCP.
- OSPF DR/BDR tie-breaking uses **Router ID**; Router ID selection should not be oversimplified to only 'highest active IP'.
- UDP checksum details differ between IPv4 and IPv6; avoid saying it is universally optional without qualification.
- TCP congestion-control behavior varies by algorithm; slow start/congestion avoidance are core concepts, but exact loss response is implementation/algorithm dependent.

Study priority: First master TCP vs UDP, OSI/TCP-IP, DNS, HTTP/HTTPS, TCP handshake, sequence/ACK, flow vs congestion control, ports, routing vs forwarding, OSPF vs RIP, and 'what happens when you type a URL'. Then revise FTP/e-mail/P2P and formulas.